



Department of Electrical and Electronic Engineering

Faculty of Engineering

University of Sri Jayewardenepura

Course	Introduction to Image Processing
Course Code	EE3253
Project	Design and implementation of a solution for a real-world applications of image processing:
Duration	4 weeks
Outcome	By the end of this practical work, a student should be: • Apply fundamental image processing techniques to solve a real-world problem. • Develop and implement an image processing application using a suitable programming environment to analyze and interpret image data. • Evaluate the performance and effectiveness of the developed solution by analyzing its accuracy, limitations, and potential improvements for practical deployment.

Instructions

Choose any one practical application of image processing (e.g., face recognition, vehicle detection, medical image analysis, license plate recognition, defect detection, traffic monitoring, plant disease detection, etc.).

Implement a code to perform the task in MATLAB or using Python. (Should be fully working code).

Report Submission

Your report should include;

Section 1: Application Selection and Problem Analysis

- Background of the selected application
- Problem statement
- Real-world significance and impact
- Technical challenges involved
- Justification for the use of selected image processing technique
- Expected system outcomes

Section 2: Methodology

- Flowchart of the processing pipeline.

- Description of algorithms/ libraries used etc.

Section 3: Implementation

- Include the evidence for your fully functional and executable code.
- You are encouraged to:
 - Compare multiple techniques where appropriate (bonus marks)
 - Use publicly available datasets

Section 4: Results

- Include the output of each step, evaluation parameters and comparison outcomes.

Section 5: Discussion and Future Improvements

- Critically discuss the limitations encountered and future enhancements.
- ***Limit your report maximum to 6 pages.***
- ***Submit the pdf version through LMS.***

Presentation and VIVA

- You are required to present your findings within 8 minutes (strictly timed presentation), followed by a 2-minute VIVA.
- Date will be informed towards the end of the project.

-End of the instruction sheet.-